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NEGOTIATION OF INTERSECTIONS BY AUTONOMOUS WORK MACHINES USING PERMISSION MANAGEMENT

Abstract

Traditionally, a management system, which grants permissions to autonomous work machines for negotiating intersections, will grant permissions in the order in which they are requested. However, this technique can often result in slowdowns and other inefficiencies in operations at the worksite. Accordingly, in disclosed embodiments, the work machines estimate entry and exit times for upcoming intersections, send these estimated times to the management system, and wait for permissions to enter the intersections. The management system determines whether conflicts exist based on the estimated times, and when a conflict exists, prioritizes the work machines involved in the conflict based on one or more factors. Then, the management system grants permissions according to the determined priorities, to thereby optimize the negotiation of intersections within the worksite.

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Background/Summary

TECHNICAL FIELD

[0001] The embodiments described herein are generally directed to negotiating intersections, and, more particularly, to the negotiation of intersections by autonomous work machines using permission management.

BACKGROUND

[0002] Autonomous work machines, such as haul trucks, at a worksite, such as a mine site, are typically managed by a central site-wide management system. The management system may manage permissions for the autonomous work machines. In particular, the management system may manage a path, through the worksite, that each work machine is permitted to drive to complete a task assigned to the work machine. The management system may divide each path into segments or “sub-paths.” When a work machine approaches the start of a sub-path, the work machine will request permission to proceed along the sub-path. In response to the request, the management system will analyze whether the sub-path is clear or occupied (e.g., either currently occupied or committed to be occupied) by another work machine. When the sub-path is clear, the management system will provide the work machine with permission to proceed along the sub-path. On the other hand, when the sub-path is occupied, the management system will withhold permission until the sub-path is clear. During this time, the work machine may have to stop and wait.

[0003] The paths of two work machines may conflict, for example, when both work machines approach the same intersection from different directions. If the work machines request permission at different times, a typical management system will simply grant permission to the work machine that requested permission first, and then, after that work machine has cleared the intersection, grant permission to the work machine that requested permission second. However, this technique does not resolve a conflict when the work machines request permission at the same time.

[0004] In addition, this technique can often result in slowdowns and other inefficiencies in operations at the worksite. For example, a first work machine that arrives at the intersection first may be forced to stop and wait for the intersection to be cleared by a second work machine that arrives at the intersection second, just because the second work machine requested permission first. This may be the case even though the first work machine could have cleared the intersection prior to the arrival of the second work machine. These inefficiencies increase dramatically as the number of intersections increases and as the number of autonomous work machines operating in the same time period increases.

[0005] There are a number of systems that are designed to manage permissions of autonomous work machines. Examples include the systems described in U.S. Pat. No. 9,672,743, issued on Jun. 6, 2017, U.S. Pat. No. 10,062,287, issued on Aug. 28, 2018, U.S. Patent Pub. No. 2023/0052077, published on Feb. 16, 2023, Japanese Patent Pub. No. 2005-225662, published on Aug. 25, 2005, and Chinese Patent No. 112419703, issued on Nov. 30, 2021. The present disclosure is directed toward overcoming one or more of the problems discovered by the inventors.

SUMMARY

[0006] In an embodiment, a method comprises, by a controller of a mobile work machine:

receiving one or more intersection locations, wherein each of the one or more intersection locations indicates a location of an intersection of two or more roads in a worksite; determining that one of the one or more intersection locations is upcoming along a path of the work machine; estimating an entry time at which the work machine will enter the intersection whose location is indicated by the one intersection location; estimating an exit time at which the work machine will exit the intersection whose location is indicated by the one intersection location; sending the estimated entry time and the estimated exit time to an external system via a wireless network; and controlling the work machine to not enter the intersection, whose location is indicated by the one intersection location, until after permission is obtained from the external system.

[0007] In an embodiment, a work machine comprises: a machine body; one or more ground-engaging members configured to move the machine body with respect to a ground; and a controller configured to receive one or more intersection locations, wherein each of the one or more intersection locations indicates a location of an intersection of two or more roads in a worksite, determine that one of the one or more intersection locations is on a path of the work machine, estimate an entry time at which the work machine will enter the intersection whose location is indicated by the one intersection location, estimate an exit time at which the work machine will exit the intersection whose location is indicated by the one intersection location, send the estimated entry time and the estimated exit time to an external system via a wireless network, and control the work machine to not enter the intersection, whose location is indicated by the one intersection location, until after permission is obtained from the external system.

[0008] In an embodiment, a method comprises, by at least one processor of a management system: receiving estimated entry times and estimated exit times from a plurality of work machines for at least one intersection; determining whether a conflict exists between two or more of the plurality of work machines based on the received estimated entry times and estimated exit times, wherein a conflict exists when the two or more work machines are estimated to be in the at least one intersection at a same time; when no conflict exists, sending permission to each of the two or more work machines; and when the conflict exists, determining priorities for the two or more work machines, and sending permission to each of the two or more work machines according to the determined priorities.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The details of embodiments of the present disclosure, both as to their structure and operation, may be gleaned in part by study of the accompanying drawings, in which like reference numerals refer to like parts, and in which:

[0010] FIG. 1 illustrates control of an autonomous work machine in a machine environment, according to an embodiment;

[0011] FIG. 2 illustrates an example processing system, by which one or more of the processes described herein, may be executed, according to an embodiment;

[0012] FIG. 3 illustrates a scenario involving autonomous work machines at a worksite, according to an example;

[0013] FIG. 4 illustrates a process for negotiating an intersection by an autonomous work machine, according to an embodiment; and

[0014] FIG. 5 illustrates a process for managing autonomous work machines, according to an embodiment.

DETAILED DESCRIPTION

[0015] The detailed description set forth below, in connection with the accompanying drawings, is intended as a description of various embodiments, and is not intended to represent the only

embodiments in which the disclosure may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of the embodiments. However, it will be apparent to those skilled in the art that embodiments of the invention can be practiced without these specific details.

[0016] In some instances, well-known structures and components are shown in simplified form for brevity of description. For clarity and ease of explanation, some surfaces and details may be omitted in the present description and figures. It should also be understood that the various components illustrated herein are not necessarily drawn to scale. In other words, the features disclosed in various embodiments may be implemented using different relative dimensions within and between components than those illustrated in the drawings.

[0017] As used herein, a reference numeral with an appended letter will be used to refer to a specific component, whereas the same reference numeral without any appended letter will be used to refer collectively to a plurality of the component or to refer to a generic or arbitrary instance of the component. In addition, the terms “respective” and “respectively” signify an association between members of a group of first items and members of a group of second items. For example, the phrase “each component A connected to a respective component B” would signify A1 connected to B1, A2 connected to B2, and so on and so forth, up to AN connected to BN.

[0018] FIG. 1 illustrates control of an autonomous work machine **110** in a machine environment **100**, according to an embodiment. Machine environment **100** may include an environment, including a worksite, in which one or more autonomous work machines **110** perform tasks associated within an industry, such as mining, construction, forestry, transportation, energy exploration, farming, or the like. For example, machine environment **100** could include a mining site in which a plurality of work machines **110** cooperate to perform a task associated with mining. However, it should be understood that disclosed embodiments are not limited to any particular environment. In addition, while work machine **110** is illustrated as a haul truck, it should be understood that work machine **110** may be any type of mobile equipment, including a wheel loader, dump truck, asphalt paver, backhoe loader, skid steer, track loader, cold planer, compactor, bulldozer, electric rope shovel, forest machine, hydraulic mining shovel, material handler, motor grader, pipe-layer, road reclaimer, telehandler, tractor-scraper, or the like, as well as ordinary road vehicles (e.g., automobiles, semi-trucks, etc.), aircraft, or maritime vessels (e.g., boats, ships, barges, etc.).

[0019] In the illustrated example, work machine **110** comprises a machine body **112** and one or more ground-engaging members **114**, configured to move machine body **112** with respect to the ground. Machine body **112** may comprise a cabin, an internal combustion engine or electric motor (e.g., driven by a battery pack), and/or the like. The cabin may comprise a seat for an operator, controls (e.g., one or more joysticks, a console comprising an integrated display with inputs, etc.), and/or the like. Ground-engaging members **114** are illustrated as wheels, but may comprise other types of components for moving machine body **112** with respect to the ground, such as tracks, rollers, and/or the like. Ground-engaging members **114** may be driven by a drivetrain, which is in turn driven by an internal combustion engine or electric motor.

[0020] Work machine **110** may comprise a controller **116**, which may be an electronic control unit (ECU). Controller **116** may be communicatively coupled (e.g., via wired or wireless communications) to one or a plurality of sensors and/or one or a plurality of subsystems of work machine **110**. Controller **116** may collect data from the sensor(s) and/or subsystem(s) of work machine **110** and process the collected data. Processing the collected data may comprise determining one or more machine states from the collected data, generating an alert based on the collected data (e.g., if a value of a parameter in the data satisfies a predefined threshold), filtering, analyzing, sorting, correlating, storing, and/or relaying the collected data, and/or the like. Controller **116** may also control the operation of one or more subsystems of work machine **110** based on the collected data and/or local or remote control commands. For example, controller **116**

may be communicatively coupled to a fuel injection system associated with the engine of work machine **110**. Controller **116** may receive a control command (e.g., increase throttle) and provide command signals to the fuel injection system, which may responsively increase the flow of fuel from a fuel supply to the engine. The engine may be connected to a torque converter that transmits power from the engine to a transmission of the drivetrain that drives ground-engaging members **114**, to thereby move work machine **110** with respect to the ground.

[0021] The sensor(s) may include any type of sensor or sensor array capable of measuring values of one or more parameters of one or more subsystems of work machine **110** and/or machine environment **100**. Examples of such parameters include, without limitation, the position of work implements, engine speed, machine speed, pressure of a fluid (e.g., fuel, oil, coolant, hydraulic fluid, etc.), flow rate of a fluid, temperature of a fluid, contamination level of a fluid, viscosity of a fluid, electric current, electric voltage, state of charge of a battery pack, fluid consumption rates, loading level, transmission output ratio, slip, grade, traction, mileage, time or mileage until or since scheduled maintenance, and/or the like. In an embodiment, the sensor(s) include a receiver of a Global Navigation Satellite System (GNSS), such as the Global Positioning System (GPS), which can be utilized to determine a location (e.g., in GPS coordinates) of work machine **110**. The sensor(s) may also include one or more cameras (e.g., regular or stereo cameras), radar systems (e.g., a Light Detecting and Ranging (LiDAR) system), and/or the like, to aid in the navigation of machine environment **100** (e.g., by providing computer vision).

[0022] Work machine **110** may comprise a communication module **118**, which may be separate or integral to controller **116**. Communication module **118** may comprise hardware and/or software that enables communication between communication module **118** and an access point of a network **120A**, which may comprise or consist of a wireless network. For example, communication module **118** may comprise or be connected to a processor, network interface, and wireless transceiver with an antenna that enables wireless communication with the base station of a cellular network. Communication module **118** may send data to a management system **130** and/or receive data from management system **130** via network **120A**. For instance, controller **116** may transmit any of the data, that are collected (e.g., from sensor(s) in work machine **110**) or derived from collected data, to management system **130** via communication module **118**, as well as receive data, such as control commands, from management system **130** via communication module **118**.

[0023] Management system **130** is one example of an external system that may host and/or execute one or more of the various functions, processes, methods, and/or software modules described herein. Management system **130** may comprise dedicated servers or may instead be implemented in a computing cloud, in which the resources of one or more servers are dynamically and elastically allocated to multiple tenants based on demand. These servers may be collocated (e.g., in a single data center) and/or geographically distributed (e.g., across multiple data centers). Management system **130** may be communicatively connected to work machine **110** via network **120A** and/or a remote terminal **140** via network **120B**. Networks **120A** and **120B** (which may be collectively referred to herein as network **120**) may be the same network, separate and distinct networks, or overlapping networks, and may include one or more cellular networks, one or more satellite networks, the Internet, one or more intranets, and/or the like. While only a single instance of work machine **110** and a single instance of remote terminal **140** are illustrated, it should be understood that management system **130** may be communicatively coupled to any number of work machines **110** and remote terminals **140**.

[0024] Management system **130** may comprise one or more web servers which host one or more websites and/or web services. In embodiments in which a website is provided, the website may comprise a graphical user interface, including, for example, one or more screens (e.g., webpages) generated in HyperText Markup Language (HTML) or other language. Management system **130** transmits or serves one or more screens of the graphical user interface in response to requests from remote terminal **140** via network **120B**. These screens may comprise a combination of content and

elements, such as text, images, videos, animations, references (e.g., hyperlinks), frames, inputs (e.g., textboxes, text areas, checkboxes, radio buttons, drop-down menus, buttons, forms, etc.), scripts (e.g., JavaScript), and/or the like.

[0025] In embodiments in which a web service is provided, management system **130** may receive requests from external systems, and provide responses in eXtensible Markup Language (XML), JavaScript Object Notation (JSON), and/or any other suitable or desired format. In such embodiments, management system **130** may provide an application programming interface (API) (e.g., implemented with a Representation State Transfer (REST) architecture) which defines the manner in which work machine **110**, remote terminal **140**, and/or other external system(s) may interact with the web service. Thus, work machine **110**, remote terminal **140**, and/or other external systems (which may themselves be servers), can define their own user interfaces, and rely on the web service to implement or otherwise provide the backend processes, methods, functionality, storage, and/or the like, described herein. For example, in such an embodiment, a client application, executing on remote terminal **140**, may interact with management system **130** to execute one or more or a portion of one or more of the various functions, processes, methods, and/or software modules described herein. In this case, the client application may generate the graphical user interface and access functionality on management system **130** via the API.

[0026] Remote terminal **140** may comprise any type of computing device capable of wired and/or wireless communication, including without limitation, desktop computers, laptop computers, tablet computers, smart phones or other mobile phones, servers, game consoles, televisions, set-top boxes, electronic kiosks, point-of-sale terminals, and/or the like. However, it is generally contemplated that remote terminal **140** is a professional workstation by which a site manager or other user may access management system **130**. For example, the user may sign in to management system **130** (e.g., using any standard authentication means, such as a username and password) to view data about the worksite, work machines **110**, and/or the like, including schedules of work machines **110**, locations and/or statuses of work machines **110**, assigned tasks and/or paths of work machines **110**, measured parameters (e.g., sensor outputs) of work machines **110**, and/or the like.

[0027] Management system **130** may operate work machines **110** autonomously, without any human operator, or semi-autonomously, with limited intervention by a human operator. During autonomous operation, management system **130** may receive data from work machines **110**, generate and send tasks for work machines **110** to complete, including assigned paths for work machines **110** to follow, generate and send control commands to work machines **110**, and/or the like. Controller **116** of each work machine **110** may send data, receive the tasks, receive the control commands, and/or the like, from management system **130**, and operate work machine **110** autonomously, based on the received information.

[0028] FIG. 2 illustrates an example processing system **200**, by which one or more of the processes described herein, may be executed, according to an embodiment. System **200** may be a wired or wireless system that is used in connection with any of the various embodiments described herein. For example, system **200** may be used as or in conjunction with one or more of the functions, processes, or methods described herein (e.g., to store and/or execute the implementing software), and may represent components of work machine **110** (e.g., controller **116**, communication module **118**, camera(s) or other sensors, etc.), management system **130**, remote terminal **140**, and/or other processing devices described herein. System **200** can be a server or any conventional personal computer, or any other processor-enabled device that is capable of wired or wireless data communication. Other computer systems and/or architectures may be also used, as will be clear to those skilled in the art.

[0029] System **200** preferably includes one or more processors **210**. Processor(s) **210** may comprise a central processing unit (CPU). Additional processors may be provided, such as a graphics processing unit (GPU), an auxiliary processor to manage input/output, an auxiliary processor to perform floating-point mathematical operations, a special-purpose microprocessor

having an architecture suitable for fast execution of signal-processing algorithms (e.g., digital-signal processor), a processor subordinate to the main processing system (e.g., back-end processor), an additional microprocessor or controller for dual or multiple processor systems, and/or a coprocessor. Such auxiliary processors may be discrete processors or may be integrated with processor **210**. Examples of processors which may be used with system **200** include, without limitation, any of the processors (e.g., Pentium™, Core i7™, Xeon™, etc.) available from Intel Corporation of Santa Clara, California, any of the processors available from Advanced Micro Devices, Incorporated (AMD) of Santa Clara, California, any of the processors (e.g., A series, M series, etc.) available from Apple Inc. of Cupertino, any of the processors (e.g., Exynos™) available from Samsung Electronics Co., Ltd., of Seoul, South Korea, any of the processors available from NXP Semiconductors N.V. of Eindhoven, Netherlands, and/or the like.

[0030] Processor **210** is preferably connected to a communication bus **205**. Communication bus **205** may include a data channel for facilitating information transfer between storage and other peripheral components of system **200**. Furthermore, communication bus **205** may provide a set of signals used for communication with processor **210**, including a data bus, address bus, and/or control bus (not shown). Communication bus **205** may comprise any standard or non-standard bus architecture such as, for example, bus architectures compliant with industry standard architecture (ISA), extended industry standard architecture (EISA), Micro Channel Architecture (MCA), peripheral component interconnect (PCI) local bus, standards promulgated by the Institute of Electrical and Electronics Engineers (IEEE), and/or the like.

[0031] System **200** preferably includes a main memory **215** and may also include a secondary memory **220**. Main memory **215** provides storage of instructions and data for programs executing on processor **210**, such as any of the software discussed herein. It should be understood that programs stored in the memory and executed by processor **210** may be written and/or compiled according to any suitable language, including without limitation C/C++, Java, JavaScript, Perl, Visual Basic, .NET, and the like. Main memory **215** is typically semiconductor-based memory such as dynamic random access memory (DRAM) and/or static random access memory (SRAM). Other semiconductor-based memory types include, for example, synchronous dynamic random access memory (SDRAM), Rambus dynamic random access memory (RDRAM), ferroelectric random access memory (FRAM), and the like, including read only memory (ROM).

[0032] Secondary memory **220** is a non-transitory computer-readable medium having computer-executable code (e.g., any of the software disclosed herein) and/or other data stored thereon. The computer software or data stored on secondary memory **220** is read into main memory **215** for execution by processor **210**. Secondary memory **220** may include, for example, semiconductor-based memory, such as programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable read-only memory (EEPROM), flash memory (block-oriented memory similar to EEPROM), and the like.

[0033] In an embodiment, I/O interface **235** provides an interface between one or more components of system **200** and one or more input and/or output devices. Example input devices include, without limitation, sensors, keyboards, touch screens or other touch-sensitive devices, joysticks, cameras, biometric sensing devices, computer mice, trackballs, pen-based pointing devices, and/or the like. Examples of output devices include, without limitation, other processing devices, cathode ray tubes (CRTs), plasma displays, light-emitting diode (LED) displays, liquid crystal displays (LCDs), printers, vacuum fluorescent displays (VFDs), surface-conduction electron-emitter displays (SEDs), field emission displays (FEDs), and/or the like. In some cases, an input and output device may be combined, such as in the case of a touch panel display (e.g., display console, smartphone, tablet computer, mobile device, etc.).

[0034] System **200** may include a communication interface **240**. Communication interface **240** allows software and data to be transferred between system **200** and an external system **245**. For example, computer software or executable code may be transferred to system **200** from a network

server (e.g., management system **130**) via communication interface **240**. Examples of communication interface **240** include a built-in network adapter, network interface card (NIC), Personal Computer Memory Card International Association (PCMCIA) network card, card bus network adapter, wireless network adapter, Universal Serial Bus (USB) network adapter, modem, a wireless data card, a communications port, an infrared interface, an IEEE 1394 fire-wire, and any other device capable of interfacing system **200** with a network (e.g., network **120**) or another computing device. Communication interface **240** preferably implements industry-promulgated protocol standards, such as Ethernet IEEE 802 standards, Fiber Channel, digital subscriber line (DSL), asynchronous digital subscriber line (ADSL), frame relay, asynchronous transfer mode (ATM), integrated digital services network (ISDN), personal communications services (PCS), transmission control protocol/Internet protocol (TCP/IP), serial line Internet protocol/point to point protocol (SLIP/PPP), and so on, but may also implement customized or non-standard interface protocols as well.

[0035] Software and data transferred via communication interface **240** are generally in the form of electrical communication signals **255**. These signals **255** may be provided to communication interface **240** via a communication channel **250**. In an embodiment, communication channel **250** may be a wired or wireless network (e.g., network **120**), or any variety of other communication links. Communication channel **250** carries signals **255** and can be implemented using a variety of wired or wireless communication means including wire or cable, fiber optics, conventional phone line, cellular phone link, wireless data communication link, radio frequency (“RF”) link, or infrared link, just to name a few.

[0036] System **200** may also include wireless communication components that facilitate wireless communication over a voice network and/or a data network. The wireless communication components, which may correspond to communication module **118** in the event that system **200** implements controller **116**, may comprise an antenna **270**, a radio **265**, and a baseband **260**. Baseband **260** is communicatively coupled with processor(s) **210**. In system **200**, RF signals are transmitted and received over the air by antenna **270** under the management of radio **265**.

[0037] Computer-executable code (e.g., computer programs, such as the disclosed software) is stored in main memory **215** and/or secondary memory **220**. Computer-executable code can also be received via communication interface **240** and/or communication module **118** (e.g., comprising baseband **260**, radio **265**, and antenna **270**) and stored in main memory **215** and/or secondary memory **220**. Such computer programs, when executed, may enable system **200** to perform the various functions of the disclosed embodiments described elsewhere herein.

[0038] In this description, the term “computer-readable medium” is used to refer to any non-transitory computer-readable storage media used to provide computer-executable code and/or other data to or within system **200**. Examples of such media include main memory **215**, secondary memory **220**, external system **245**, and any peripheral device communicatively coupled with communication interface **240** (including a network information server or other network device). These non-transitory computer-readable media are means for providing software and/or other data for utilization in system **200**.

[0039] FIG. **3** illustrates a scenario involving autonomous work machines **110** at a worksite, according to an example that will be used to described embodiments. In the illustrated scenario, the worksite comprises a plurality of roads **310**, illustrated as roads **310A**, **310B**, and **310C**, with a plurality of intersections **320**, illustrated as intersection **320AB** of roads **310A** and **310B** and intersection **320AC** of roads **310A** and **310C**. A plurality of work machines **110**, illustrated as work machines **110A**, **110B**, and **110C**, are operational within the worksite during the same time period.

[0040] In this scenario, work machine **110A** is on a path along road **310A**, and work machine **110B** is on a path along road **310B**. Work machines **110A** and **110B** are simultaneously approaching intersection **320AB**. Work machine **110A** is closest to intersection **320AB** and requests permission to cross intersection **320AB** before work machine **110B**. However, work machine **110B** is traveling

at a much higher speed than work machine **110A**, such that work machine **110B** will reach intersection **320AB** before work machine **110A**. For ease of description, it will be assumed that the paths of work machine **110A** and work machine **110B** are both straight through intersection **320AB**. However, it should be understood that one or both of the paths could alternatively turn left or right at intersection **320AB**.

[0041] Traditionally, the work machine **110** that first requested permission to cross intersection **320AB** would receive permission to cross intersection **320AB** first. Thus, in this case, traditionally, work machine **110A** would receive permission to cross intersection **320AB** before work machine **110B**. This would require work machine **110B**, which reaches intersection **320AB** before work machine **110A**, to decelerate and stop at boundary **325B1** of intersection **320AB**, wait for work machine **110A** to clear intersection **320AB**, and then accelerate and proceed through intersection **320AB**. However, this is inefficient, since work machine **110B** could have cleared intersection **320AB** before work machine **110A** reached intersection **320AB**, such that no work machine **110** needed to stop. In other words, the time required for work machine **110B** to decelerate to a stop, wait, and accelerate from the stop represents an unnecessary slowdown.

[0042] FIG. **4** illustrates a process **400** for negotiating an intersection **320** by an autonomous work machine **110**, according to an embodiment. Process **400** may be implemented in software and/or hardware by controller **116** (e.g., as embodied in a system **200**). Process **400** may execute continuously for as long as work machine **110** is operational, and end, along with other functions of work machine **110**, when work machine **110** is shutdown (i.e., operation of work machine **110** is terminated). While process **400** is illustrated with a certain arrangement and ordering of subprocesses, process **400** may be implemented with fewer, more, or different subprocesses and a different arrangement and/or ordering of subprocesses. In addition, it should be understood that any subprocess, which does not depend on the completion of another subprocess, may be executed before, after, or in parallel with that other independent subprocess, even if the subprocesses are described or illustrated in a particular order.

[0043] In subprocess **405**, controller **116** receives one or more, and generally a plurality of, intersection locations. Each intersection location indicates the location of an intersection **320** of two or more roads **310** in the worksite. In an embodiment, controller **116** receives these intersection location(s) from management system **130**, via communication module **118** and network **120A**. The locations of roads **310** and/or intersections **320** may change, and/or roads **310** and/or intersections **320** may be added or removed, in some worksites, such as mining sites. Accordingly, management system **130** may periodically send the intersection location(s) to controller **116** of each work machine **110** according to a fixed time interval, whenever there is a change to an intersection location, whenever a new intersection location is added, whenever an existing intersection is removed, and/or the like. Additionally or alternatively, management system **130** may send the intersection location(s) to a work machine **110** in response to a request from that work machine **110**. Whenever controller **116** receives intersection location(s) from management system **130**, controller **116** may store and/or update the intersection location(s) in persistent memory, such as secondary memory **220**.

[0044] In an embodiment, the intersection location(s) may be provided in the form of a representation (e.g., map) of the worksite, with the location of each intersection **320** in the representation of the worksite tagged by a marker that identifies the intersection **320** (e.g., using a unique identifier of intersection **320**). More generally, each landmark in the worksite may be tagged with a marker that identifies the landmark (e.g., using a unique identifier of the landmark). Of particular relevance to disclosed embodiments, the landmarks comprise one or more intersections **320**. Examples of other types of landmarks, include, without limitation, bridges, bends in a road **310**, waiting (e.g., parking) areas, loading areas, dumping areas, fueling or charging stations, and/or the like. Landmarks may also include the end of fixed distance intervals, such as mile markers. In the illustrated scenario, management system **130** may send, to each of work machines **110A**, **110B**,

and **110C**, a representation of the worksite with at least the locations of intersection **320AB** and intersection **320AC** tagged by a marker identifying the respective intersection.

[0045] In subprocess **410**, controller **116** determines whether or not a task has been assigned to work machine **110**. In an embodiment, work machine **110** is assigned tasks by management system **130**. In particular, management system **130** may optimize production at the worksite by managing the tasks performed by each work machine **110**. When management system **130** assigns a new task to a work machine **110**, management system **130** may send the assigned task to the work machine **110** via network **120A**. Work machine **110** may receive the assigned task, which may include a path along one or more roads **310** and through one or more intersections **320** within the worksite, and begin operation according to the assigned task. When work machine **110** has no assigned task (i.e., “No” in subprocess **410**), process **400** may proceed to subprocess **415**. Otherwise, when work machine **110** has an assigned task (i.e., “Yes” in subprocess **410**), process **400** may proceed to subprocess **420**.

[0046] In subprocess **415**, controller **116** may wait for a new task. In particular, controller **116** may retain work machine **110** in, or switch work machine **110** to, a waiting mode. In the waiting mode, controller **116** may control work machine **110** to stop where it is and wait for a task, travel to a waiting area where it can stop and wait for a task (e.g., a parking area that is out of the way of other work machines **110**), travel to a fueling station to be refueled or to a charging station to be charged, perform one or more default tasks, or the like.

[0047] In subprocess **420**, controller **116** may determine the path for the assigned task. The path may be provided by management system **130** (e.g., when sending the task). Alternatively, controller **116** may determine the path based on information provided for the assigned task by management system **130**. As mentioned above, the path may be divided into sub-paths, and controller **116** may request permission to proceed along each sub-path in advance of reaching the start of each sub-path. However, disclosed embodiments are directed to the negotiation of intersections. Accordingly, for the sake of simplicity, other aspects of permission management will be omitted from the present disclosure. It should be understood that the disclosed embodiments may be utilized within a larger permission-management architecture.

[0048] In the illustrated scenario, each of work machines **110A**, **110B**, and **110C** have already been assigned respective tasks. Each work machine **110** has already determined a respective path. In particular, the path of work machine **110A** comprises a straight sub-path along road **310A** and through intersections **320AB** and **320AC**, the path of work machine **110B** comprises a straight sub-path along road **310B** and through intersection **320AB**, and the path of work machine **110C** comprises a straight sub-path along road **310C** and through intersection **320AC**.

[0049] In subprocess **425**, controller **116** may control work machine **110** to proceed along the path that was determined in subprocess **420**. In particular, controller **116** may control the engine (e.g., internal combustion engine, electric motor, etc.) and steering system of work machine **110** to drive autonomously along a road **310**, according to the determined path. In the illustrated scenario, controller **116** of work machine **110A** controls work machine **110A** to drive along road **310A** towards intersection **320AB**, controller **116** of work machine **110B** controls work machine **110B** to drive along road **310B** towards intersection **320AB**, and controller **116** of work machine **110C** controls work machine **110C** to drive along road **310C** towards intersection **320AC**.

[0050] In subprocess **430**, controller **116** may determine whether or not the assigned task is complete. The assigned task may be determined to be complete when work machine **110** has reached a final destination along the path determined in subprocess **420**, and potentially performed some action at the final destination (e.g., a loading or unloading action). When determining that the assigned task is not complete (i.e., “No” in subprocess **430**), process **400** may proceed to subprocess **435**. Otherwise, when determining that the assigned task is complete (i.e., “Yes” in subprocess **430**), process **400** may return to subprocess **410** to await a new task.

[0051] In subprocess **435**, controller **116** may determine whether or not work machine **110** is

approaching an intersection **320**. In particular, controller **116** may continually monitor the current location of work machine **110** (e.g., using a GPS receiver) and the locations of tagged landmarks, including intersections **320**, that are upcoming along the path that was determined in subprocess **420**. When determining that an intersection location is upcoming along the path of work machine **110** (i.e., “Yes” in subprocess **435**), process **400** may proceed to subprocess **440**. Otherwise, when determining that no intersection location is upcoming along the path of work machine **110** (i.e., “No” in subprocess **435**), process **400** may return to subprocess **425**, such that work machine **110** continues along the path.

[0052] In subprocess **435**, controller **116** of work machine **110** may determine either only the next intersection **320** along the path, all intersections **320** that remain along the path, or up to a predefined number of consecutive next intersections **320** along the path. In the illustrated scenario, controllers **116** of work machine **110A** and work machine **110B** both determine that intersection **320AB** is upcoming, and controller **116** of work machine **110C** determines that intersection **320AC** is upcoming. In addition, controller **116** of work machine **110A** may also determine that intersection **320AC** is upcoming after intersection **320AB**.

[0053] In subprocess **440**, controller **116** may estimate an entry time at which work machine **110** will enter the intersection **325** whose location is indicated by each intersection location determined to be upcoming along the path of work machine **110**. Each entry time may be estimated by calculating a time (e.g., an absolute time, time duration from the current time, or the like), at which work machine **110** will arrive at the start of intersection **325**, based on the current speed of work machine **110**, the current distance between work machine **110** and the start of intersection **325**, the machine profile of work machine **110**, the geometry of road **310** (e.g., uphill vs. downhill, straight vs. curved, etc.), the state of work machine **110** (e.g., with or without payload), the presence of any obstacles (e.g., determined by computer-vision, using cameras or other sensors, onboard work machine **110**), and/or the like.

[0054] In the illustrated scenario, controller **116** of work machine **110A** estimates the time at which work machine **110A** will arrive at a first boundary **325A1** of intersection **320AB**, controller **116** of work machine **110B** estimates the time at which work machine **110B** will arrive at a first boundary **325B1** of intersection **320AB**, and controller **116** of work machine **110C** estimates the time at which work machine **110C** will arrive at a first boundary **325C1** of intersection **320AC**. In addition, controller **116** of work machine **110A** may estimate the time at which work machine **110A** will arrive at a first boundary **325A3** of intersection **320AC**.

[0055] In subprocess **445**, controller **116** may estimate an exit time at which work machine **110** will exit the intersection **325** whose location is indicated by each intersection location determined to be upcoming along the path of work machine **110**. Each exit time may be estimated by calculating a time (e.g., an absolute time, time duration from the current time, or the like), at which work machine **110** will clear the end of intersection **325**, based on the current speed of work machine **110**, the current distance between work machine **110** and the end of intersection **325**, the machine profile of work machine **110** (e.g., including the length of work machine **110**), the geometry of road **310** (e.g., uphill vs. downhill, straight vs. curved, etc.), the state of work machine **110** (e.g., with or without payload), the presence of any obstacles (e.g., determined by computer-vision, using cameras or other sensors, onboard work machine **110**), and/or the like.

[0056] In the illustrated scenario, controller **116** of work machine **110A** estimates the time at which work machine **110A** will clear a second boundary **325A2** of intersection **320AB**, controller **116** of work machine **110B** estimates the time at which work machine **110B** will clear a second boundary **325B2** of intersection **320AB**, and controller **116** of work machine **110C** estimates the time at which work machine **110C** will clear a second boundary **325C2** of intersection **320AC**. In addition, controller **116** of work machine **110A** may estimate the time at which work machine **110A** will clear a second boundary **325A4** of intersection **320AC**. It should be understood that embodiments will work just as well for paths that turn through an intersection **320**. For example, if the path of

work machine **110A** includes a left turn through intersection **320AB**, instead of a straight sub-path through intersection **320AB**, controller **116** of work machine **110A** would estimate the time at which work machine **110A** will clear second boundary **325B2**, instead of second boundary **325A2**. [0057] In subprocess **450**, controller **116** may send the entry time, estimated in subprocess **440**, and the exit time, estimated in subprocess **445**, to management system **130**, via communication module **118** and network **120A**. Management system **130** may receive the estimated entry and exit times for a plurality of work machines **110**, operating at the worksite, and determine the optimal order of permissions for work machines **110** to cross intersections **325**. Then, management system **130** may send permissions to work machines **110**, over network **120A**, in the optimal order. For each intersection **320**, after management system **130** has sent a permission to one work machine **110**, management system **130** may wait until that work machine **110** clears intersection **320** before granting any subsequent permission. The operation of management system **130** is described in greater detail elsewhere herein.

[0058] In subprocess **455**, controller **116** determines whether or not permission to cross intersection **320** has been received. Permission may be received from management system **130**, via communication module **118** and network **120A**. When permission to cross intersection **320** is received (i.e., “Yes” in subprocess **455**), process **400** may return to subprocess **425**, such that work machine **110** continues along the path. Otherwise, when no permission to cross intersection **320** has been received (i.e., “No” in subprocess **455**), process **400** may proceed to subprocess **460**.

[0059] In subprocess **460**, controller **116** may wait until permission to cross intersection **325** is received. In particular, controller **116** may retain work machine **110** in, or switch work machine **110** to, a waiting mode. In the waiting mode, controller **116** may control work machine **110** (e.g., an engine and/or braking system of work machine **110**) to reduce the speed of work machine **110**, potentially to a full stop, in advance of intersection **325**. In some cases, it may not be necessary for work machine **110** to come to a full stop, for example, when permission is received in subprocess **455** before work machine **110** has come to a full stop. In any case, controller **116** controls work machine **110** to not enter intersection **320** until after permission is obtained from management system **130**.

[0060] In an embodiment, controller **116** may minimize stopping, to thereby reduce fuel or battery consumption. When a work machine **110** is forced to decelerate to a full stop, wait, and then accelerate to speed again, substantial fuel or battery charge is consumed. Thus, in an embodiment, as work machine **110** approaches an intersection **320** while having not yet received permission to enter intersection **320**, controller **116** may reduce the speed of work machine **110** with a goal of not reaching intersection **320** before permission has been received, to thereby prevent the need to stop at intersection **320**, and thereby reduce fuel or battery consumption. Controller **116** may reduce the speed of work machine **110** based on the distance between the current location of work machine **110** and the location of intersection **320**, an estimated time at which work machine **110** will receive permission to cross intersection **320**, and/or the like. The time at which work machine **110** will receive permission to cross intersection **320** may be provided by management system **120** over network **120A**.

[0061] Management system **120** may determine the estimated time at which each work machine **110** will receive permission to cross intersection **320**, based on the estimated exit times received for each work machine **110** with respect to intersection **320**. As will be discussed elsewhere herein, management system **120** may prioritize work machines **110** when a conflict exists for an intersection **320**. Management system **120** may determine the estimated time at which each work machine **110** will receive permission to cross intersection **320** based on the estimated exit times for each work machine **110** having higher priority than that work machine **110**.

[0062] In the illustrated scenario, in subprocess **450**, controller **116** of each work machine **110A**, **110B**, and **110C** may send the estimated entry and exit times to management system **130** over network **120A** via their respective communication modules **118**. Management system **130** may

determine that work machine **110B** can cross intersection **320AB** before work machine **110A** even reaches intersection **320AB**. Thus, management system may firstly grant permission to work machine **110B** to cross intersection **320AB** (e.g., received in subprocess **455** by controller **116** of work machine **110B**), and then, once work machine **110B** has cleared intersection **320AB**, secondly grant permission to work machine **110A** to cross intersection **320AB** (e.g., received in subprocess **455** by controller **116** of work machine **110A**). In this case, neither work machine **110A** nor **110B** may have to decelerate or stop. However, if the estimated times turn out to be incorrect (e.g., because work machine **110A** and/or **110B** changes speed, noise in the estimates, etc.), such that work machine **110A** reaches intersection **320AB** before work machine **110B** has entered or cleared intersection **320AB**, work machine **110A** will stop and wait before crossing boundary **325A1**, since it will not yet have received permission. Thus, safety is maintained. It should be understood that work machines **110A** and **110C** may negotiate intersection **320AC** in the same manner.

[0063] In an embodiment that minimizes stopping, controller **116** of work machine **110A** may determine that it will not receive permission before it reaches intersection **320AB** at its current speed. For example, the estimated exit time for work machine **110B** may precede the estimated entry time for work machine **110A**. In this case, management system **130** may communicate the estimated time, at which work machine **110A** will be granted permission to cross intersection **320AB**, to controller **116** via network **120A**. In response, controller **116** may reduce the speed of work machine **110A**, such that the estimated entry time of work machine **110A**, at first boundary **325A1** of intersection **320AB**, is at or after work machine **110B** has cleared intersection **320AB**. This prevents the need for work machine **110A** to have to stop at intersection **320AB**. After work machine **110B** has cleared intersection **320AB**, controller **116** may return work machine **110A** to its previous speed. This temporary reduction in speed may also resolve any conflict between work machine **110A** and work machine **110C** at intersection **320AC**. Otherwise, controllers **116** of work machine **110A** and/or work machine **110C** may perform the same technique for intersection **320AC**, to thereby prevent any need for work machine **110A** or **110C** to stop.

[0064] FIG. 5 illustrates a process **500** for managing autonomous work machines **110**, according to an embodiment. Process **500** may be implemented in software and/or hardware by management system **130** (e.g., as embodied in a system **200**). While process **500** is illustrated with a certain arrangement and ordering of subprocesses, process **500** may be implemented with fewer, more, or different subprocesses and a different arrangement and/or ordering of subprocesses. In addition, it should be understood that any subprocess, which does not depend on the completion of another subprocess, may be executed before, after, or in parallel with that other independent subprocess, even if the subprocesses are described or illustrated in a particular order.

[0065] In subprocess **505**, management system **130** provides each work machine **110** with one or more, and generally a plurality of, intersection locations in the worksite. Management system **130** may periodically send the intersection location(s) to each work machine **110** according to a fixed time interval, whenever there is a change to an intersection location, whenever a new intersection location is added, whenever an existing intersection is removed, and/or the like. Additionally or alternatively, management system **130** may send the intersection location(s) to a work machine **110** in response to a request from that work machine **110**. As discussed elsewhere herein, the intersection location(s) may be sent as tagged landmarks in a representation (e.g., map) of the worksite.

[0066] In subprocess **510**, management system **130** determines whether or not any estimated times have been received from work machine(s) **110**. In particular, management system **130** may receive estimated entry times and estimated exit times from each of one or more work machines **110**. Of particular relevance to disclosed embodiments, management system **130** may receive estimated entry times and estimated exit times from a plurality of work machines **110** for each of one or more intersections **320**. When estimated times are received (i.e., “Yes” in subprocess **510**), process **500** may proceed to subprocesses **515** and/or **545**. Otherwise, when estimated times are not received

(i.e., “No” in subprocess **510**), process **500** may continue to wait for new estimated times to be received.

[0067] Subprocesses **515** and **545** represent entry points to discrete, parallel processes within process **500**. The first discrete process, comprising subprocesses **515-540**, identifies and resolves conflicts, if any, between work machines **110** at intersections **320**. The second discrete process, comprising subprocesses **545-555**, optimizes traffic within the worksite. The estimated entry and exit times from work machines **110** for intersection(s) **320** may be used as input to both of these discrete processes. In addition, the second discrete process may utilize other times, estimated by and received from work machines **110**, such as the estimated arrival time of a work machine **110** at another type of landmark.

[0068] In subprocess **515**, management system **130** may determine whether or not a conflict exists between two or more work machines **110** based on the received estimated entry times and estimated exit times. A conflict exists when two or more work machines **110** are estimated to be in an intersection **320** at the same time. In particular, a time period in which each work machine **110** is estimated to be in each of one or more intersections **320** may be determined based on the estimated entry and exit times for that work machine **110**. For example, if the estimated times represent absolute times, the time period that work machine **110** is estimated to be in an intersection **320** is the time period beginning with the estimated entry time and ending with the estimated exit time. A conflict exists if the time periods, in which two or more work machines **110** are estimated to be in the same intersection **320**, overlap. A buffer may be incorporated into the overlap (e.g., by adding a buffer period to the start and/or end of each time period) to provide for error tolerance in the estimations and ensure that each work machine **110** is fully clear of an intersection **320** before granting permission to another work machine **110**. When no conflict exists (i.e., “No” in subprocess **515**), process **500** may proceed to subprocess **520**. Otherwise, when a conflict exists (i.e., “Yes” in subprocess **515**), process **500** may proceed to subprocess **525**.

[0069] In subprocess **520**, a permission is sent to each of the two or more work machines **110**. In particular, management system **130** may send permissions to work machines **110** over network **120A**. Notably, when there is no conflict, the order in which the permissions are sent may not matter. Thus, management system **130** may send the permissions serially in any order or in parallel. In any case, work machines **110** will proceed along their respective paths, including through intersection(s) **320**.

[0070] In subprocess **525**, management system **130** may prioritize the work machines **110** involved in the conflict that was identified in subprocess **515**. Management system **130** may determine the priorities of the work machines **110**, involved in the conflict, based on one or more factors. For example, management system **130** may calculate a value of the task to which each of work machines **110** is assigned based on one or more parameters (e.g., weighted parameters), and prioritize work machines **110**, relative to each other, based on the calculated values. The value of the task assigned to a work machine **110** may represent the cost of delaying that work machine **110** from completing its task. In this case, work machines **110** whose tasks have a higher value or cost should be prioritized over work machines **110** whose tasks have a lower value or cost. The one or more parameters may comprise, for example, for each of work machines **110**, at least one of a cost of a final destination of the path of work machine **110**, an estimated arrival time of work machine **110** at the final destination of the path, an indication of whether or not work machine **110** is hauling a load, an estimated time duration that work machine **110** will have to wait to enter intersection **320**, a fuel level or state of charge of work machine **110**, and/or the like.

[0071] After determining the priorities of work machines **110**, permissions may be sent to each of the two or more work machines **110**, involved in the conflict, according to the determined priorities. In particular, an iteration of subprocesses **530-540** may be performed until all of the work machines **110**, involved in the conflict, have been sent permissions. At a high level, in each iteration, permission is sent to the remaining work machine **110** with the highest priority, and when

another work machine **110** remains to receive permission, management system **130** waits until the most recent work machine **110** to receive permission has cleared intersection **320** before sending another permission.

[0072] In subprocess **530**, management system **130** may send a permission to the one of the remaining work machine(s) **110** having the highest priority, according to the priorities that were determined in subprocess **525**. It should be understood that the remaining work machine(s) **110** are those that are involved in the conflict and have not yet been sent permission. To track which work machines **110** have not yet been sent permission, representations of work machines **110**, involved in the conflict, may be ordered according to the priorities of work machines **110**, and queued such that the work machine **110** with the highest priority is at the front of the queue and the work machine **110** with the lowest priority is at the back of the queue. Then, in each iteration of subprocess **530**, the representation of work machine **110** at the front of the queue is removed and the respective work machine **110** is sent a permission to cross intersection **320**.

[0073] It is possible that, during the described negotiation process, a new work machine **110** could approach the intersection and report its respective estimated entry and exit times. If the new work machine **110** is in conflict with the remaining work machine(s) **110**, as determined by the estimated entry and exit times reported by the new work machine **110**, the new work machine **110** can be added to the queue of work machine(s) **110**. In this case, the priorities of the remaining work machines **110**, including the newly added work machine **110**, may be redetermined in subprocess **525**, and the queue may be reordered in subprocess **530** according to the redetermined priorities.

[0074] In subprocess **535**, management system **130** may determine whether or not intersection **320** is clear. Intersection **320** may be determined to be clear when the work machine **110**, which received permission in the most recent iteration of subprocess **530**, has fully crossed the boundary representing the end of intersection **320** along the path of that work machine **110**. In an embodiment, a buffer period may be added to ensure that work machine **110** is fully clear of intersection **320**. Controller **116** of each work machine **110** may notify management system **130**, over network **120A**, when work machine **110** has cleared intersection **320**, and management system **130** may determine that intersection **320** is clear when the notification is received from the most recent work machine **110** to be sent permission. When determining that intersection **320** is clear (i.e., “Yes” in subprocess **535**), process **500** may proceed to subprocess **540**. Otherwise, when determining that intersection **320** is not clear (i.e., “No” in subprocess **535**), process **500** may continue to wait for work machine **110** to clear intersection **320**.

[0075] In subprocess **540**, management system **130** determines whether or not another work machine **110**, involved in the conflict, remains to be considered. In an embodiment in which a queue is used, it may be determined that a work machine **110** remains to be considered when the queue is not empty, and determined that no work machine **110** remains to be considered when the queue is empty. When determining that a work machine **110** remains to be considered (i.e., “Yes” in subprocess **540**), process **500** may return to subprocess **530** to consider the next work machine **110**, according to the determined priorities. Otherwise, when determining that no work machine **110** remains to be considered (i.e., “No” in subprocess **540**), process **500** may return to subprocess **510**.

[0076] In subprocess **545**, management system **130** may record the estimated times that were received in subprocess **510**. For example, management system **130** may store the estimated entry times and estimated exit times, from potentially a plurality of work machines **110** and for potentially a plurality of intersections **320**, in persistent memory (e.g., secondary memory **220**) of management system **130**. Management system **130** may also record other estimated times, such as estimated arrival times of work machines **110** at other types of landmarks. This record represents an estimated schedule of traffic within the worksite for at least a portion of the paths of work machines **110**. Management system **130** may also receive and store the states of work machines **110**, such that the record also represents the state of each work machine **110**.

[0077] In subprocess **550**, management system **130** may estimate travel times for the remaining

portions of the paths of work machines **110**, based on the recorded estimated times, including the received estimated entry times and estimated exit times. In other words, management system **130** may estimate, for each work machine **110**, the future travel time required for that work machine **110** to complete its assigned task, based on the estimated entry and/or exit times of one or more intersections **320** along the path of that work machine **110**, the estimated time of arrival of that work machine **110** at one or more other landmarks, a current speed of work machine **110**, the remaining distance from that work machine **110** to the destination of the path of that work machine **110**, the machine profile of work machine **110**, the geometry of road(s) **310** on the remaining portion of the path (e.g., uphill vs. downhill, straight vs. curved, etc.), the state of work machine **110** (e.g., with or without payload), the presence of any obstacles, and/or the like.

[0078] In subprocess **555**, traffic on the worksite may be optimized based on the travel times of the work machines **110** that were estimated in subprocess **550**. In particular, management system **130** may execute an optimization model that utilizes the estimated travel times as an input. The optimization model may optimize the travel times, task assignments, the respective speeds of work machines **110**, the paths of work machines **110**, and/or the like. The optimization model may comprise a machine-learning algorithm that minimizes or maximizes an objective function representing a target to be optimized (e.g., minimize a total travel time, maximize a total value, minimize a total cost, etc.). The output of optimization model may be new or updated tasks and/or paths that may be sent by management system **110** to one or more work machines **110**, to thereby optimize traffic within the worksite, for example, to achieve target travel times for work machines **110**. In other words, the traffic may be optimized by adjusting one or more paths of work machine(s) **110**, adjusting one or more tasks of work machine(s) **110**, adjusting the speed of one or more work machines **110**, stopping one or more work machines **110**, reassigning one or more work machines **110**, and/or the like.

[0079] Embodiments have been primarily described herein using a central management system **130**, for example, as the external system from which permissions to cross intersections **320** are received by work machines **110**. However, in an alternative embodiment, the external system may be another work machine **110**. In this case, obtaining permission from the external system, for example, in subprocess **455**, may comprise negotiating permission with the other work machine **110**. In particular, autonomous work machines **110** may implement peer-to-peer communications that enable each work machine **110** to communicate with any other work machine **110** within a certain range (e.g., 10 meters, 50 meters, 100 meters, 500 meters, etc.). Thus, each work machine **110** may send estimated entry and/or exit times to other work machines **110** within range, and, if a conflict exists, the work machines **110**, involved in the conflict, may negotiate permissions using peer-to-peer communications. Such negotiation can be performed in the absence of any management system **130** or in the event that management system **130** becomes unavailable (e.g., due to failure of management system **130**, network **120A**, or the like).

Industrial Applicability

[0080] Traditionally, a management system, which grants permissions to autonomous work machines for negotiating intersections, will grant permissions in the order in which they are requested. However, this technique can often result in slowdowns and other inefficiencies in operations at the worksite.

[0081] Accordingly, in disclosed embodiments, work machines **110** estimate entry and exit times for upcoming intersections **320** along their paths, and send these estimated entry and exit times to an external system, such as management system **130**, which may determine whether or not a conflict exists in an intersection **320**. If a conflict exists, the external system may prioritize the work machines **110**, involved in the conflict, and send permissions according to the determined priorities, thereby providing more efficient intersection negotiation that can be optimized so as to minimize the consumption of fuel or battery charge.

[0082] Each work machine **110** may wait for permission to enter any intersection **320**. In an

embodiment, a work machine **110**, which does not expect to receive permission before reaching an intersection **320**, may reduce its speed well in advance of the intersection **320**, in order to avoid having to stop at the intersection **320**. This may eliminate or otherwise reduce the consumption of fuel or battery charge by the deceleration to a full stop, the waiting while stopped, and the acceleration from a full stop, by work machine **110**.

[0083] In an embodiment, the external system may be a management system **130** that determines whether conflicts exist based on the estimated entry and exit times. When a conflict exists, management system **130** may prioritize the work machines involved in the conflict based on one or more factors. Then, the management system **130** grants permissions according to the determined priorities, to thereby optimize the negotiation of intersections within the worksite.

[0084] In addition, management system **130** may utilize the estimated times to estimate travel times for the remaining portions of the paths of work machines **110**. Management system **130** may utilize the estimated travel times to optimize traffic within the worksite, thereby improving production efficiency and further reducing fuel or battery consumption by work machines **110**.

[0085] It will be understood that the benefits and advantages described above may relate to one embodiment or may relate to several embodiments. Aspects described in connection with one embodiment are intended to be able to be used with the other embodiments. Any explanation in connection with one embodiment applies to similar features of the other embodiments, and elements of multiple embodiments can be combined to form other embodiments. The embodiments are not limited to those that solve any or all of the stated problems or those that have any or all of the stated benefits and advantages.

[0086] The preceding detailed description is merely exemplary in nature and is not intended to limit the invention or the application and uses of the invention. The described embodiments are not limited to usage in conjunction with a particular type of machine. Hence, although the present embodiments are, for convenience of explanation, depicted and described as being implemented in a work machine, it will be appreciated that it can be implemented in various other types of mobile equipment, and in various other systems and environments. Furthermore, there is no intention to be bound by any theory presented in any preceding section. It is also understood that the illustrations may include exaggerated dimensions and graphical representation to better illustrate the referenced items shown, and are not considered limiting unless expressly stated as such.

Claims

1. A method comprising, by a controller of a mobile work machine: receiving one or more intersection locations, wherein each of the one or more intersection locations indicates a location of an intersection of two or more roads in a worksite; determining that one of the one or more intersection locations is upcoming along a path of the work machine; estimating an entry time at which the work machine will enter the intersection whose location is indicated by the one intersection location; estimating an exit time at which the work machine will exit the intersection whose location is indicated by the one intersection location; sending the estimated entry time and the estimated exit time to an external system via a wireless network; and controlling the work machine to not enter the intersection, whose location is indicated by the one intersection location, until after permission is obtained from the external system.
2. The method of claim 1, wherein controlling the work machine comprises reducing a speed of the work machine.
3. The method of claim 1, wherein controlling the work machine comprises stopping the work machine.
4. The method of claim 1, wherein the worksite is a mining site.
5. The method of claim 1, wherein the one or more intersection locations are a plurality of intersection locations.

6. The method of claim 1, wherein the external system is another work machine, and wherein obtaining permission from the external system comprises negotiating permission with the other work machine.
7. The method of claim 1, wherein the external system is a management system, and wherein obtaining permission from the external system comprises receiving permission from the management system.
8. The method of claim 7, wherein the method further comprises, by at least one processor of the management system: receiving estimated entry times and estimated exit times from a plurality of work machines for at least one intersection; determining whether a conflict exists between two or more of the plurality of work machines based on the received estimated entry times and estimated exit times, wherein a conflict exists when the two or more work machines are estimated to be in the at least one intersection at a same time; when no conflict exists, sending permission to each of the two or more work machines; and when the conflict exists, determining priorities for the two or more work machines, and sending permission to each of the two or more work machines according to the determined priorities.
9. The method of claim 8, wherein determining priorities for the two or more work machines comprises: calculating a value of a task to which each of the two or more work machines is assigned, based on one or more parameters; and prioritizing the two or more work machines, relative to each other, based on the calculated values.
10. The method of claim 9, wherein the one or more parameters comprise, for each of the two or more work machines, at least one of a cost of a destination of the path of the work machine, an estimated arrival time of the work machine at the destination of the path, an indication of whether or not the work machine is hauling a load, an estimated time duration that the work machine will have to wait to enter the at least one intersection, or a fuel or energy level of the work machine.
11. The method of claim 8, wherein sending permission to each of the two or more work machines according to the determined priorities comprises, until all of the two or more work machines have been sent permissions, performing an iteration of: sending permission to one of the two or more work machines having a highest priority and which has not yet been sent permission; and when another of the two or more work machines remains to receive permission, waiting until the one work machine has cleared the at least one intersection before sending another permission.
12. The method of claim 7, wherein the method further comprises, by at least one processor of the management system: receiving estimated entry times and estimated exit times from a plurality of work machines for a plurality of intersections; estimating travel times for remaining portions of the paths of the plurality of work machines based on the received estimated entry times and estimated exit times; and optimizing traffic within the worksite based on the estimated travel times.
13. The method of claim 12, wherein optimizing traffic comprises adjusting one or more of the paths of the plurality of work machines.
14. The method of claim 12, wherein optimizing traffic comprises adjusting one or more tasks of the plurality of work machines.
15. A work machine comprising: a machine body; one or more ground-engaging members configured to move the machine body with respect to a ground; and a controller configured to receive one or more intersection locations, wherein each of the one or more intersection locations indicates a location of an intersection of two or more roads in a worksite, determine that one of the one or more intersection locations is on a path of the work machine, estimate an entry time at which the work machine will enter the intersection whose location is indicated by the one intersection location, estimate an exit time at which the work machine will exit the intersection whose location is indicated by the one intersection location, send the estimated entry time and the estimated exit time to an external system via a wireless network, and control the work machine to not enter the intersection, whose location is indicated by the one intersection location, until after permission is obtained from the external system.

16. A method comprising, by at least one processor of a management system: receiving estimated entry times and estimated exit times from a plurality of work machines for at least one intersection; determining whether a conflict exists between two or more of the plurality of work machines based on the received estimated entry times and estimated exit times, wherein a conflict exists when the two or more work machines are estimated to be in the at least one intersection at a same time; when no conflict exists, sending permission to each of the two or more work machines; and when the conflict exists, determining priorities for the two or more work machines, and sending permission to each of the two or more work machines according to the determined priorities.

17. The method of claim 16, wherein determining priorities for the two or more work machines comprises: calculating a value of a task to which each of the two or more work machines is assigned, based on one or more parameters; and prioritizing the two or more work machines, relative to each other, based on the calculated values.

18. The method of claim 17, wherein the one or more parameters comprise, for each of the two or more work machines, at least one of a cost of a destination of a path of the work machine, an estimated arrival time of the work machine at the destination of the path, an indication of whether or not the work machine is hauling a load, an estimated time duration that the work machine will have to wait to enter the at least one intersection, or a fuel or energy level of the work machine.

19. The method of claim 16, wherein sending permission to each of the two or more work machines according to the determined priorities comprises, until all of the two or more work machines have been given permissions, performing an iteration of: sending permission to one of the two or more work machines having a highest priority and which has not yet received permission; and when another of the two or more work machines remains to receive permission, waiting until the one work machine has cleared the at least one intersection before sending another permission.

20. The method of claim 16, wherein the method further comprises, by the at least one processor of the management system: estimating travel times for remaining portions of paths of the plurality of work machines based on the received estimated entry times and estimated exit times; and optimizing traffic within the worksite based on the estimated travel times.
